



Empower Security and Compliance with Azure Policy



Gregor Reimling

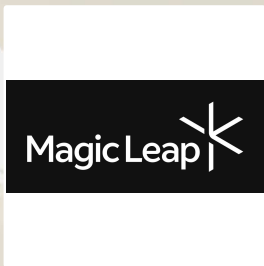
**aMS Aachen
16.11.2023**



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About “Gregor Reimling”



Focus

Azure Governance, Security
and IaaS

Certifications

Cloud Security Architect, MVP
for MS Azure & Security

From

Cologne, Germany

Hobbies

Family, Community,
Worldtraveler

My Blog

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Agenda

				
Governance ^{NEW}	Security	Resiliency	Monitoring	Automate
Proactively apply policies and optimize cloud spend	Industry leading Security with Advanced Threat Protection	High availability and protection for VMs, apps and data	Deep operational insights with rich intelligence	Powerful scripting, configuration and update management

How it works

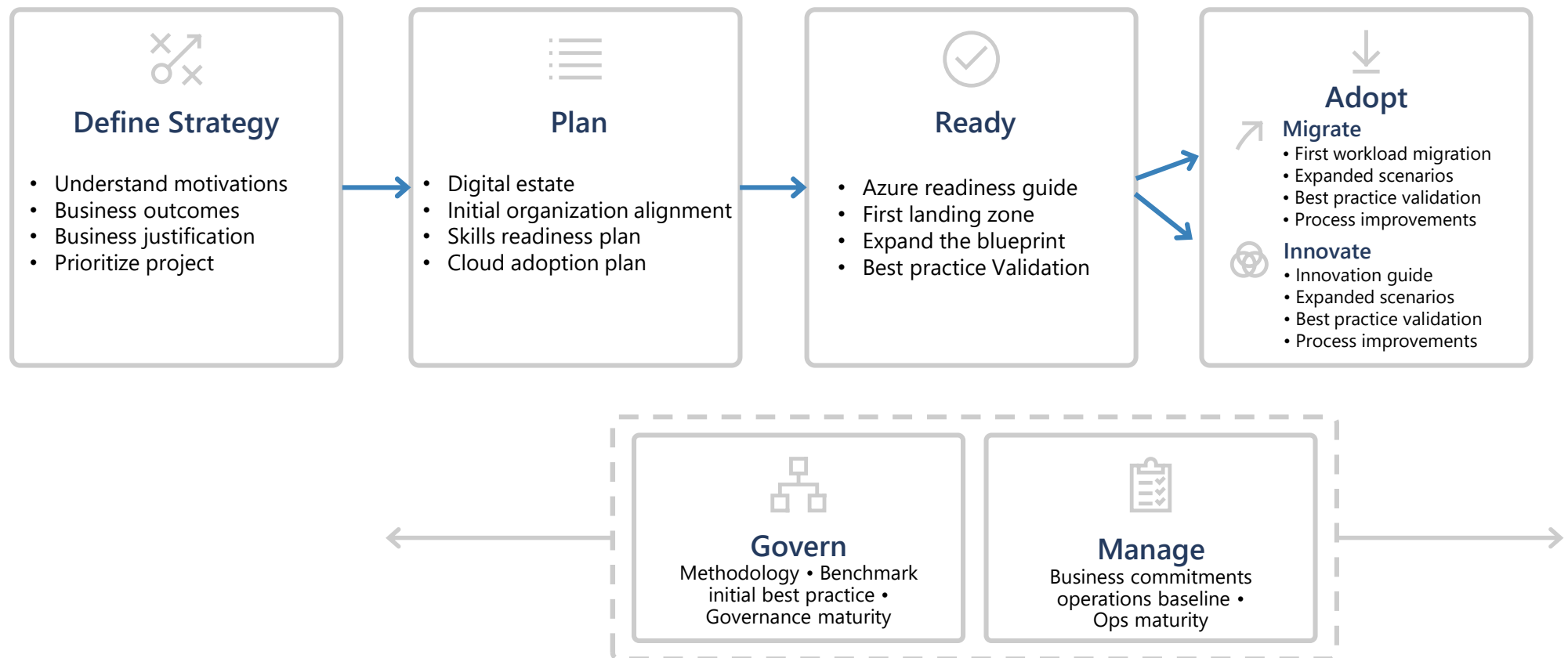
Policies at scale

Recommended Policies

Automation and Remediation

MS Defender for Cloud

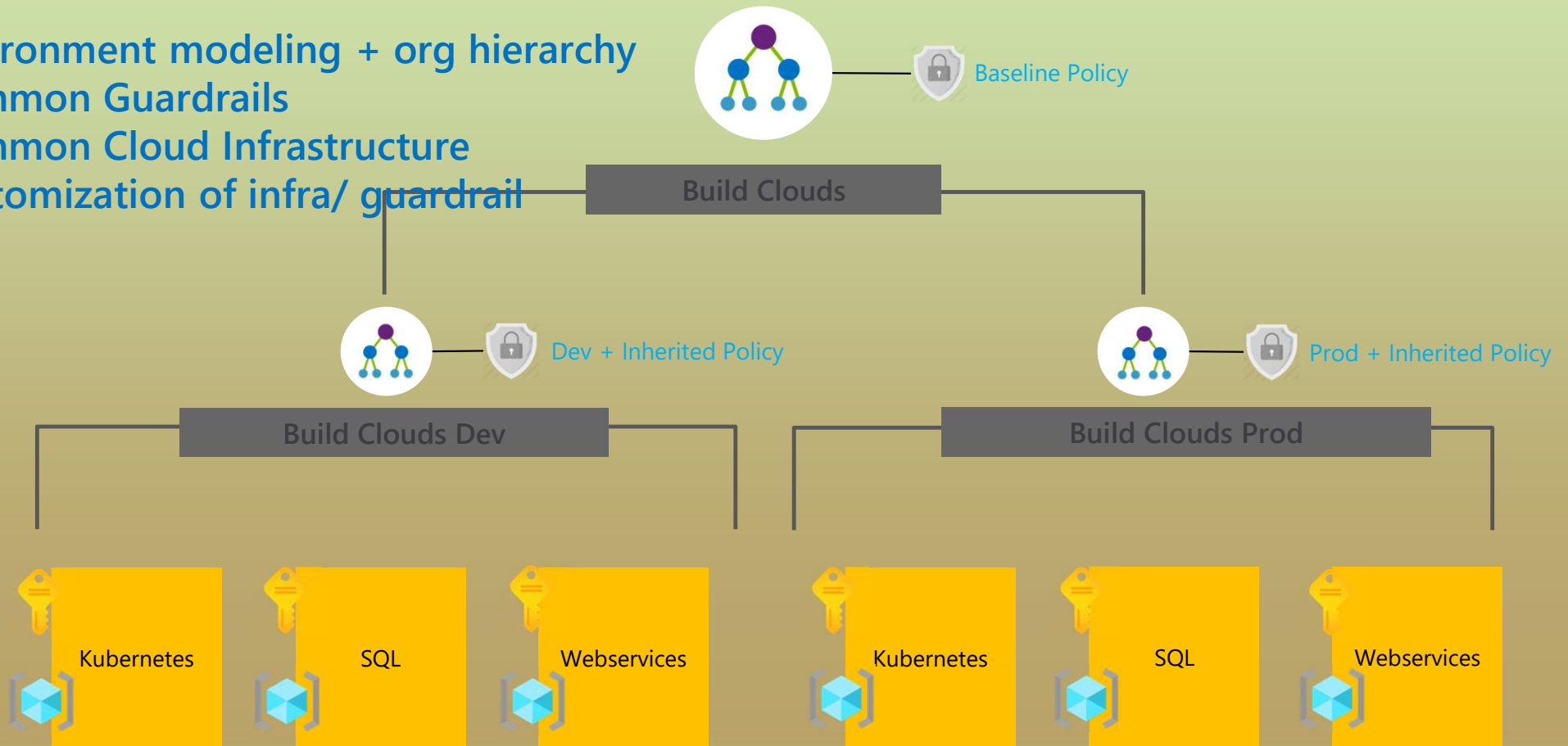
Microsoft Cloud Adoption Framework for Azure



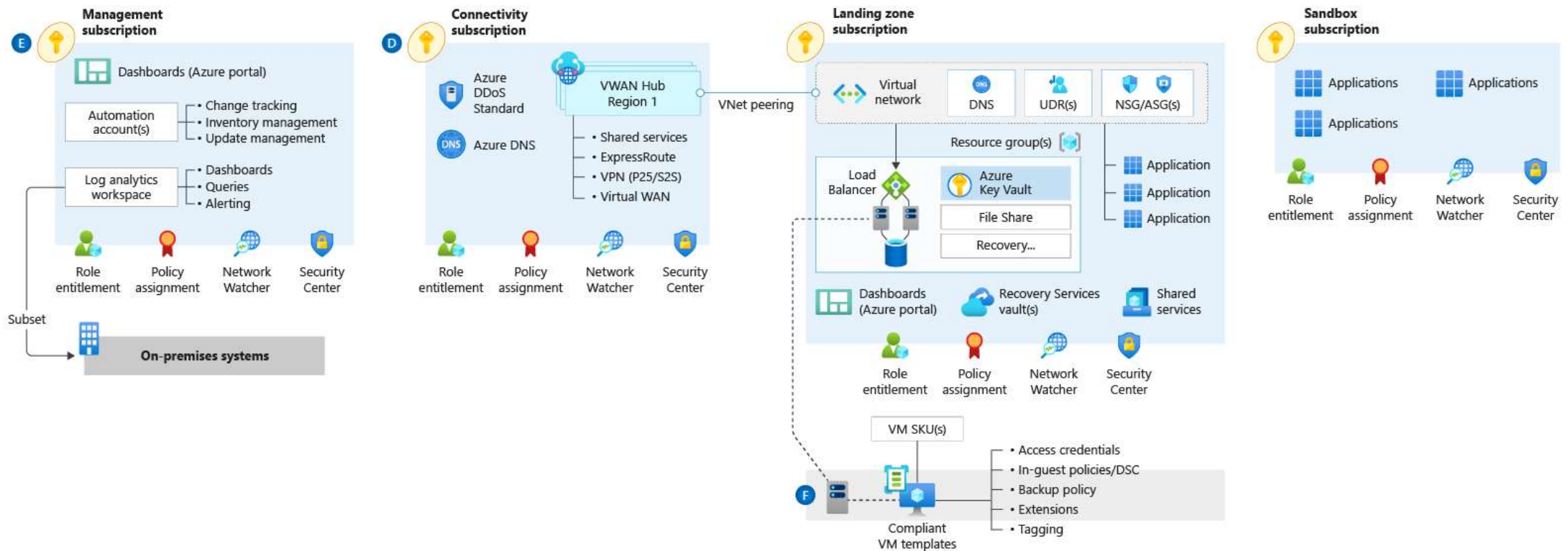
<https://azure.microsoft.com/en-us/cloud-adoption-framework/>

Management Groups

Environment modeling + org hierarchy
Common Guardrails
Common Cloud Infrastructure
Customization of infra/ guardrail

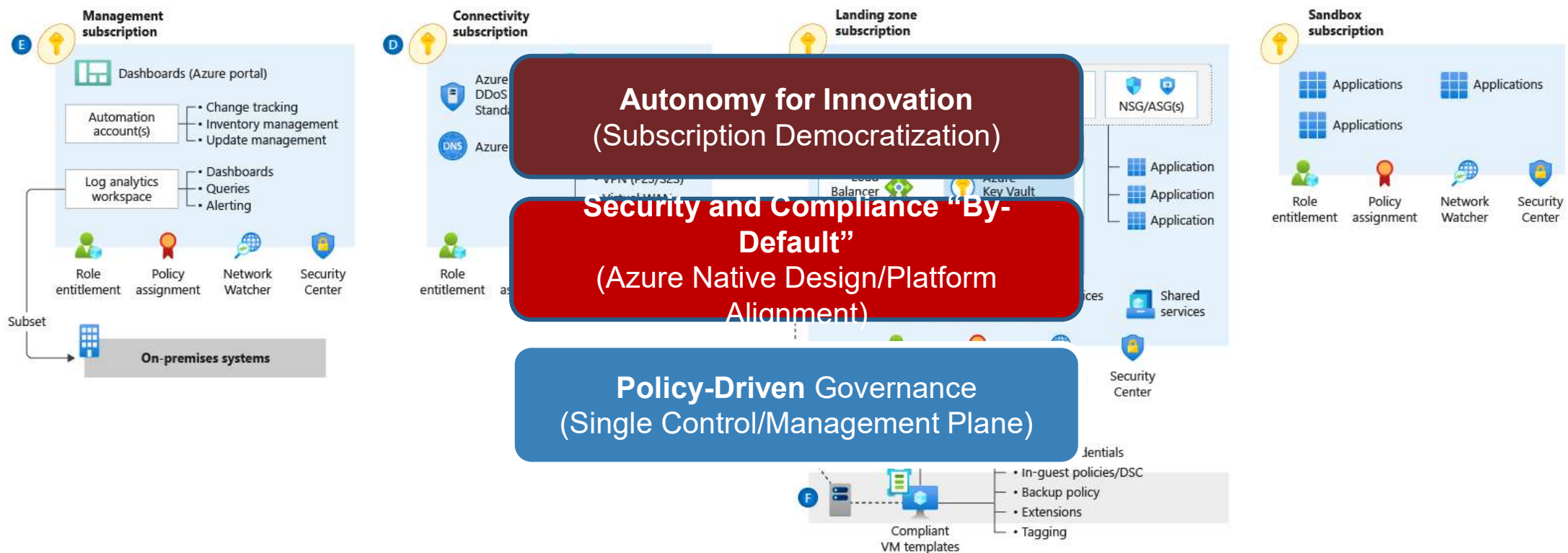


Enterprise-Scale - Design Principles



Source: [GitHub-Repo "Azure/Enterprise-Scale"](#) - Deploy Enterprise-Scale foundation

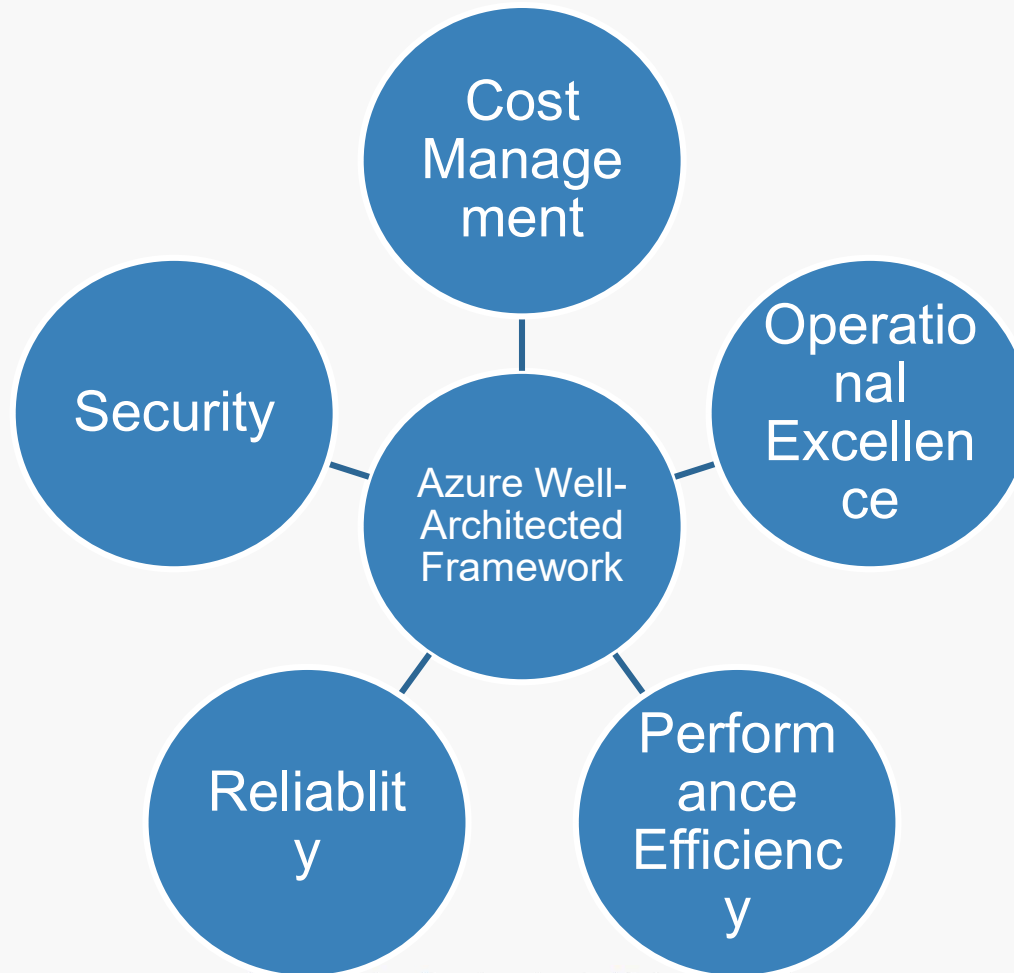
Enterprise-Scale - Design Principles



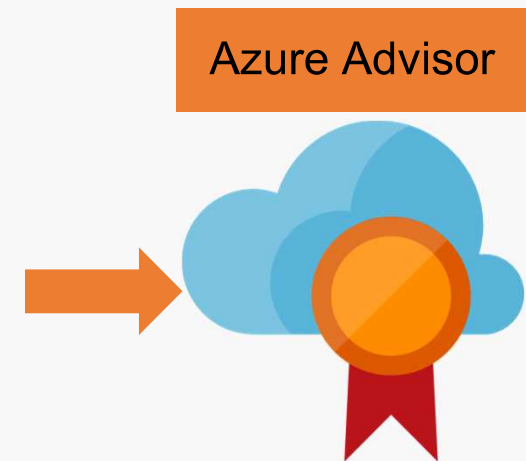
Source: [GitHub-Repo "Azure/Enterprise-Scale"](#) - Deploy Enterprise-Scale foundation

Well-architected Framework

"The Azure Well-Architected Framework is a set of guiding tenets that can be used to improve the quality of a workload."



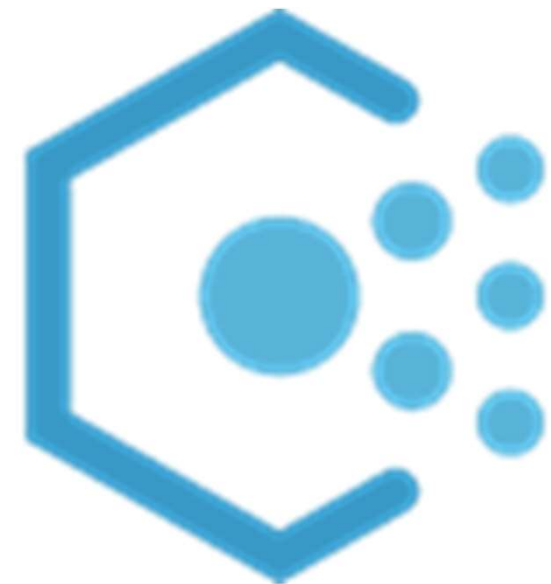
<https://docs.microsoft.com/en-us/azure/architecture/framework/>





What are Azure Policy

- Part of CAF to support WAF
- Create, assign and manage policies
- Enforce rules to ensure your ressources are compliant
- Focus on ressource properties for new and existing deployments
- A definition is a set of conditions in audit or deny mode
- An assignment is a policy definition placed on a specific scope
- An initiative is a collection of policies



Azure Policy



- Turn on built-in policies or build custom ones for all resource types
- Real-time policy evaluation and enforcement
- Periodic & on-demand compliance evaluation
- VM In-Guest Policy (**NEW**)

Enforcement & Compliance



- Apply policies to a Management Group with control across your entire organization
- Apply multiple policies and & aggregate policy states with policy initiative
- Exclusion Scope

Apply policies at scale



- Real time remediation
- Remediation on existing resources (**NEW**)

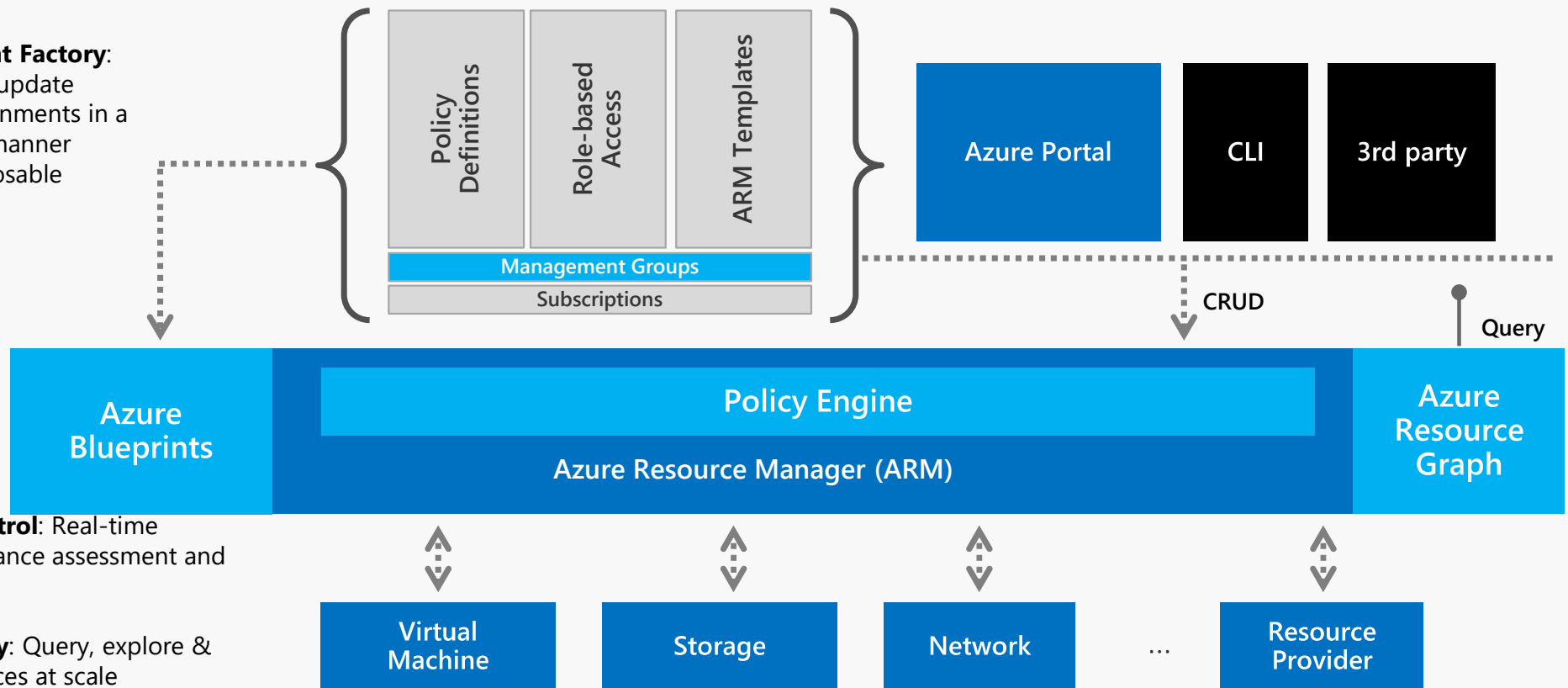
Remediation

Azure Governance Architecture

providing control over the cloud environment, without sacrificing developer agility

1. Environment Factory:

Deploy and update cloud environments in a repeatable manner using composable artifacts



2. Policy-based Control: Real-time enforcement, compliance assessment and remediation at scale

3. Resource Visibility: Query, explore & analyze cloud resources at scale

Leverage built-in initiative & policies



Security

Azure Security Center

Guest Config

baselines

Key Vault certificate

NSG rules

AKS & AKS Engine

RBAC role

assignment



Regulatory Compliance

NIST SP 800-53 R4

ISO 27001:2013

CIS

PCI v3.2.1:2018

FedRAMP Moderate

Canada Federal PBMM

SWIFT CSP-CSCF v2020

UK Official and UK NHS

IRS 1075



Tags

Require specified tag

Add or replace a tag

Inherit a tag from the

RG

Append a tag



Resource standardization

Allowed/ not allowed

RP

Allowed locations

Naming convention

Back up VMs

Allowed images for

AKS



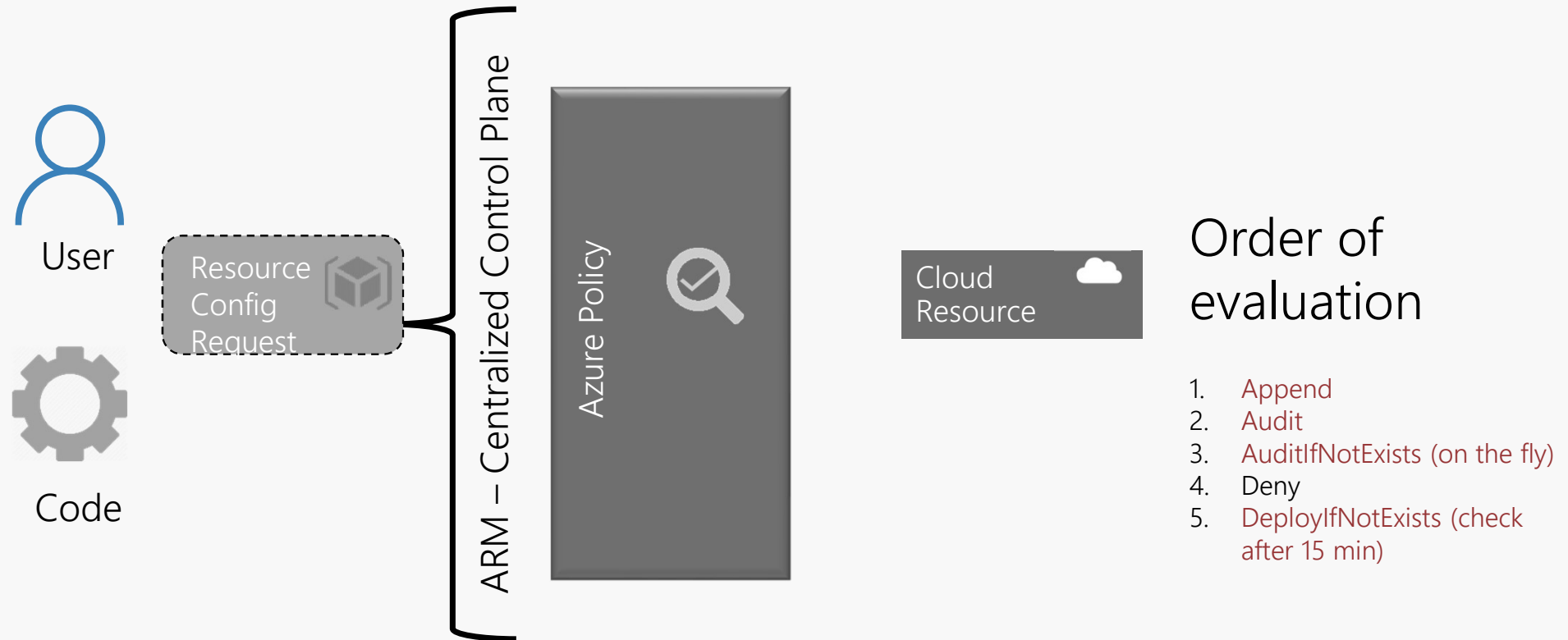
Cost

Allowed VM SKUs

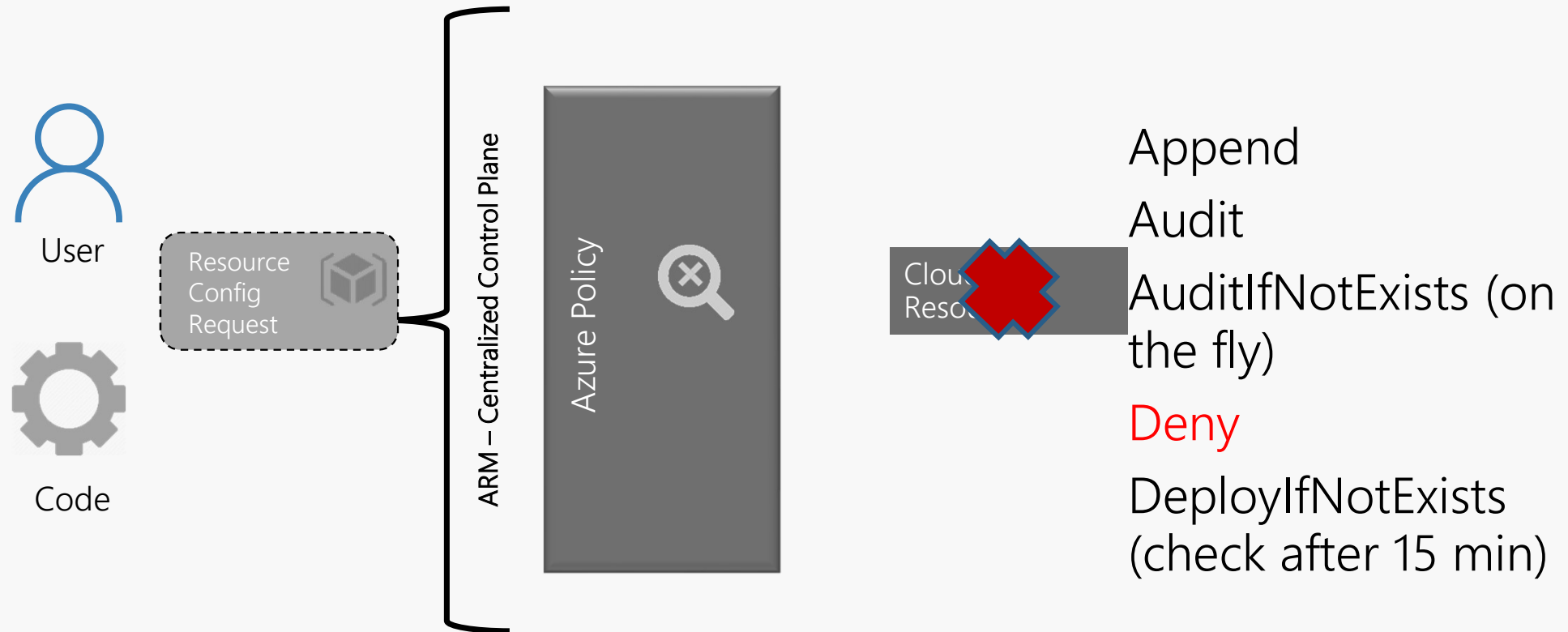
Allowed Storage

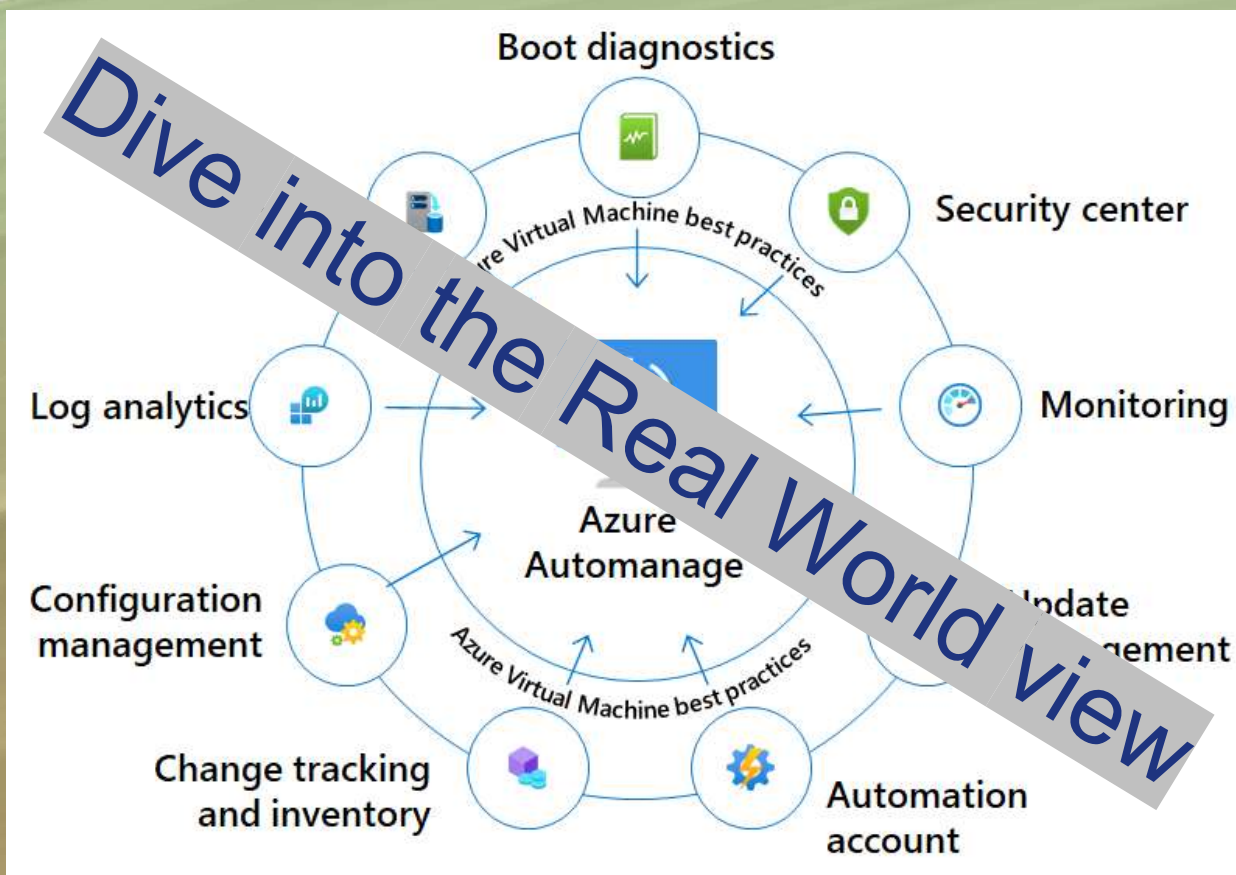
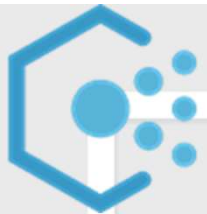
SKUs

How does it work?



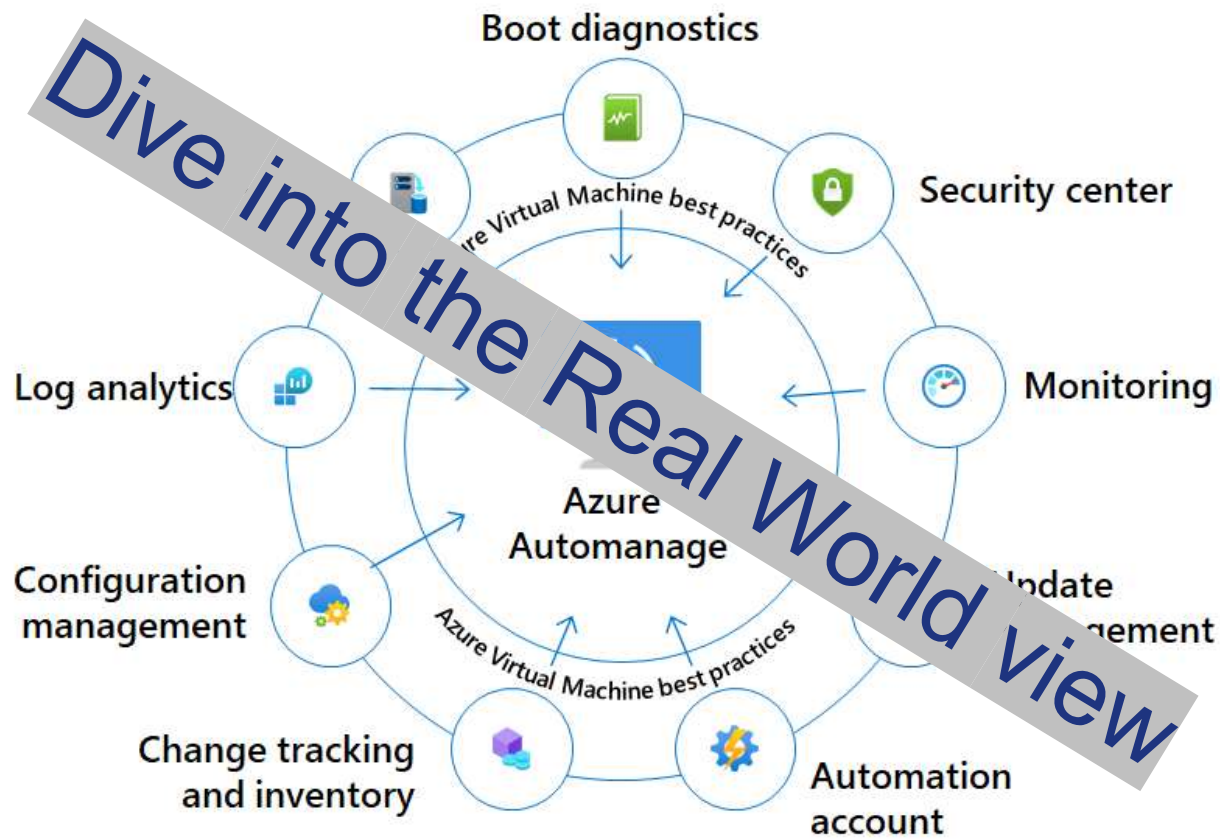
How does it work?





Azure Policy Definition structure

```
{  "properties": {  
    "mode": "all",  
    "parameters": {  
        "allowedLocations": {  
            "type": "array",  
            "metadata": { "description": "The list of locations that can be specified when deploying resources",  
                "strongType": "location",  
                "displayName": "Allowed locations" }, "defaultValue": [ "westus2" ]  
        } },  
    "displayName": "Allowed locations",  
    "description": "This policy enables you to restrict the locations your organization can specify when deploying resources.",  
    "policyRule": {  
        "if": {  
            "not": {  
                "field": "location", "in": "[parameters('allowedLocations')]"  
            } },  
        "then": { "effect": "deny"
```



Microsoft Defender for Cloud



A SERVICE TO
STRENGTHEN
YOUR
SECURITY
POSTURE



AVAILABLE IN
TWO TIERS –
ASC BASIC
AND AZURE
DEFENDER



BASIC -> FREE
– ACTIVATED
BY DEFAULT
FOR ALL
SUBSCRIPTION
S



BASED ON AN
SECURITY
SCORE –
SCOPE BASED



AVAILABLE
FOR ALL
WORKLOADS
(SERVER,
CONTAINER,
SQL, IOT AND
HYPERCONVERGENT)

Microsoft Defender for Cloud



Strengthen security posture

Cloud security posture management

Secure Score

Policies and compliance



Protect against threats

For
servers

For cloud native
workloads

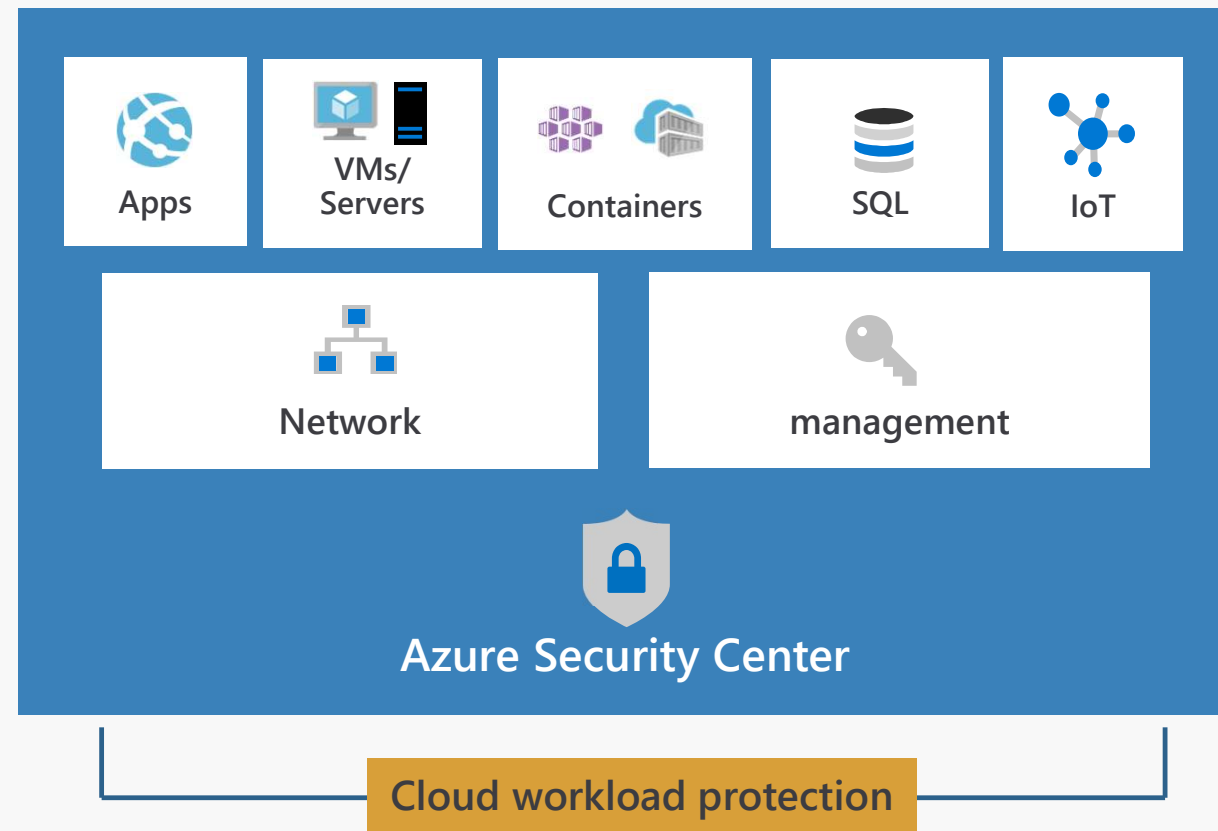
For
databases
and storage



Get secure faster

Protect your workloads

- Detect & block advanced malware and threats for Linux and Windows Servers on any cloud
- Protect cloud-native services from threats
- Protect data services against malicious attacks
- Protect your Azure IoT solutions with near real time monitoring
- Service layer detections: Azure network layer and Azure management layer (ARM)



Microsoft Defender for Cloud

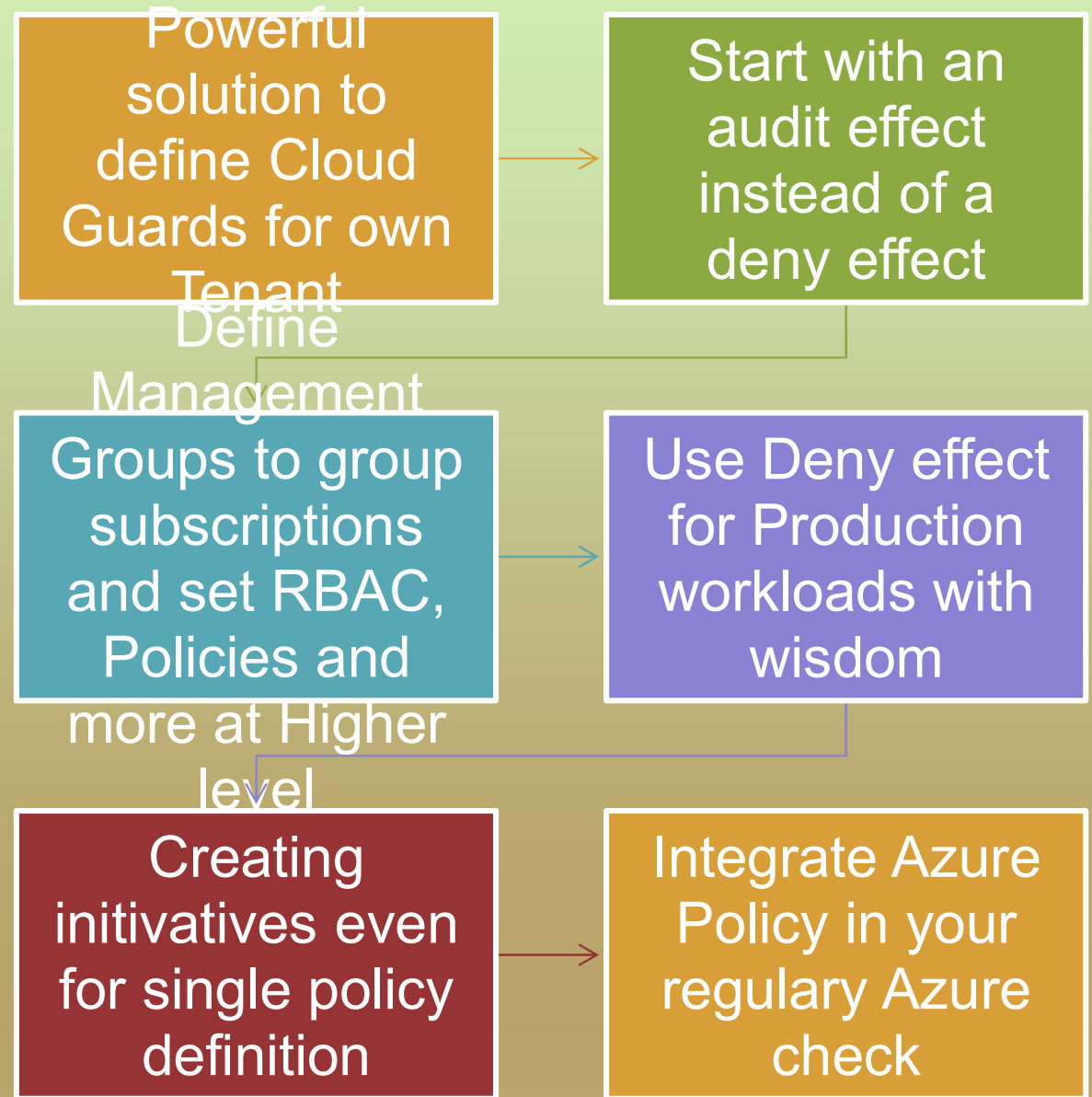


DEMO

How it works together

- All Azure Security Center recommendations based on Azure Policy
- Secure score is result of Azure Policy settings
- Recommendation is also a result of Azure Policy
- All Azure Policy are defined in Compliance mode
- Azure Policy settings for ASC will firstly applied when Subscription is created

Azure Policy Recap



Microsoft Defender for Cloud

- Start with ASC to get a Security Overview
- Use ASC to strengthen your infrastructure
- Check the status in ASC regularly
- Create own security policies for secure score
- Use ASC to proof your infrastructure
- Integrate Azure Policy and ASC in your regular Azure check

Links

<https://docs.microsoft.com/en-us/azure/governance/policy/overview>

<https://docs.microsoft.com/en-us/azure/governance/policy/>

<https://docs.microsoft.com/en-us/azure/governance/policy/tutorials/create-and-manage>

<https://www.reimling.eu/2019/03/azure-management-groups-und-blueprints-ueberblick-und-einrichtung-teil-1/>

<https://github.com/Azure/azure-policy/>

<https://aka.ms/SecurityCommunity>

<https://docs.microsoft.com/en-us/azure/security-center/>

<https://docs.microsoft.com/en-us/azure/security-center/security-center-policy-definitions>

<https://docs.microsoft.com/en-us/azure/security-center/custom-security-policies>

<https://blog.azureandbeyond.com/2020/04/24/mastering-azure-security-my-latest-adventure/>

<https://github.com/Azure/Azure-Security-Center>

<https://techcommunity.microsoft.com/t5/azure-security-center/weekly-secure-score-progress-report/ba-p/2159354>

<https://github.com/Azure/Azure-Security-Center/tree/master/Secure%20Score/SecureScoreOverTimeReport>

Training: <https://aka.ms/ascninja>

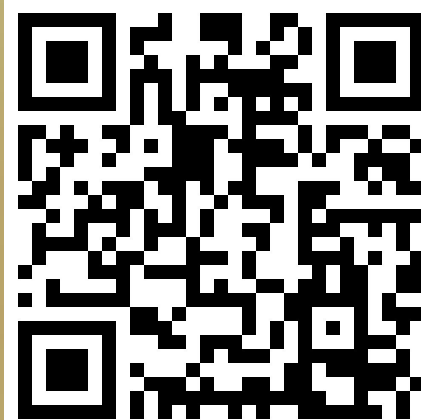
Videos: <https://aka.ms/ascinthevideo>

ASC Lab: <https://aka.ms/aslabs>

[Jesse \(JSON\) Loudon \(jloudon.com\)](#)



Thank You



Blog
<https://www.Reimling.eu>



Contact



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